

PCB Passives: Capacitors, Resistors & Diodes

1. Capacitor Types, Polarity & Decoupling Practices

Capacitors store electrical energy electrostatically and filter noise on circuit supply rails.

Capacitor Type	Typical Values	Polarity & Typical Usage
Electrolytic (Aluminum)	1 uF to 4700 uF	POLARIZED. Long lead is (+), stripe is (-). Bulk power filtering.
Ceramic (Multilayer MLCC)	10 pF to 10 uF	NON-POLARIZED. Symmetrical. High-frequency noise bypass.
Tantalum	0.1 uF to 220 uF	POLARIZED. Highly sensitive to overvoltage. Low ESR bypass.
Film (Polyester/Poly)	1 nF to 1 uF	NON-POLARIZED. Audio, timing filters, precision resonators.

FATAL HAZARD: Reverse Polarity on Electrolytic Capacitors. Connecting negative voltage to the positive lead of an aluminum electrolytic capacitor initiates rapid electrolysis of the aluminum oxide dielectric, producing hydrogen gas, case rupture, and catastrophic explosion. Always verify polarity stripe before powering up!

2. Resistor Color Codes & Power Derating

4-Band Resistor Decoding Rule: • Band 1 = 1st Significant Digit | Band 2 = 2nd Significant Digit | Band 3 = Multiplier (10^N) | Band 4 = Tolerance (Gold = +-5%, Silver = +-10%). • Color Values: Black(0), Brown(1), Red(2), Orange(3), Yellow(4), Green(5), Blue(6), Violet(7), Gray(8), White(9).

Example: Brown-Black-Red-Gold = 1 - 0 * 10^2 = 1,000 Ohms (1 kOhm) +-5%.

Power Dissipation Law: $P = I^2 * R = V^2 / R$. Standard axial resistors are rated for 0.25W (1/4 Watt). Derating guideline: Never operate a resistor above 60% of its rated wattage (e.g., maximum 0.15W on a 1/4W resistor) to prevent thermal degradation and resistance drift.

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3. Diodes & Semiconductor Protection Devices

- 1N4007 Standard Silicon Rectifier: Maximum Repetitive Reverse Voltage = 1000V, Forward Current = 1.0A. Forward Voltage Drop $V_f = 0.7V$. Primary use: 50/60 Hz AC line rectification and DC reverse-polarity protection.
- 1N4148 High-Speed Switching Diode: Reverse Recovery Time $t_{rr} = 4 \text{ ns}$, Max Current = 200 mA. Primary use: High-speed digital logic, clipping, small signal detection.
- 1N5819 Schottky Barrier Diode: Forward Voltage Drop $V_f = 0.35V$ at 1A, ultra-fast switching. Primary use: High-efficiency DC-DC buck converters and inductive flyback suppression.
- Zener Diodes (e.g. 1N4733A 5.1V): Operates in reverse breakdown region to clamp voltage. Must always use series current-limiting resistor to prevent thermal destruction.

4. Circuit Troubleshooting & Multimeter Testing Protocol

- Diode Mode Test: Red probe on Anode, Black probe on Cathode (side with silver stripe). Silicon diode reads $\sim 0.5V$ to $0.7V$. Schottky reads $\sim 0.2V$ to $0.4V$. Reverse bias should read 'OL' (Open Loop). If reading $0.00V$ in both directions, diode is shorted.
- In-Circuit Resistor Testing: Resistance measurements taken with power OFF. Note that parallel components in the circuit will cause the multimeter to read a LOWER resistance than the nominal value. To confirm an open resistor, desolder one lead.
- Blown Electrolytic Capacitor Signs: Bulging or domed aluminum top vent, brown electrolytic fluid crust around base, or high ESR causing excessive ripple on oscilloscope.